Doorbell System for Deaf People

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***Abstract* —** *Innovation has always helped solve human challenges, making life easier and more efficient. From electric cars to washing machines, technology has improved various aspects of our daily lives. It has also played a crucial role in assisting people with disabilities.*

This project aims to help deaf individuals respond to important non-verbal sounds, such as a doorbell, allowing them to live more independently. The system consists of a wireless module (NRF) that sends signals when the doorbell is pressed. A wearable device with LED lights and a vibrating motor will alert the user. The entire system is controlled by an Arduino unit, ensuring efficient and reliable communication between the doorbell and the wearable device.

Keywords *— Arduino, Wireless module, vibrating motor, LED, Buzzer*

I. Introduction

Technology has always assisted individuals with disabilities in living improved lives. From assisting a blind individual to read to aiding those who are unable to speak in communicating, innovation has turned the impossible into a reality. Nowadays, technology is so sophisticated that we can monitor our fitness, track our sleep, and do many more things just with a basic mobile app. These advancements have facilitated the lives of individuals of all ages and also provided confidence to those who are helpless because of their disabilities.

Not everybody has the money to spend on costly medical treatment for hearing loss. But with the help of assistive technology, they can get over most obstacles. For instance, speech therapy enables individuals to enhance their voice so that they can communicate better. Several organizations provide this training to children and adults. In the same way, intelligent gloves can translate sign language into speech, enabling deaf individuals to communicate freely with others. The gloves employ sensors and a Bluetooth chip to detect hand movements and translate them into words.

Living without one of our senses is very challenging. Just imagine walking on a crowded street in Mumbai without hearing even a single sound—how risky and infuriating would it be? This thought encouraged us to develop a smart gadget that assists deaf individuals in reacting to doorbells. Our project will inform them whenever anyone rings the doorbell, through a wearable gadget with lights and vibrations. This easy yet efficient solution can ease their daily lives significantly.

II. LITERATURE REVIEW

The evolution of assistive technology for persons with hearing loss has attracted much interest in recent decades, especially in the context of doorbell alert systems. Conventional audio-based doorbell systems do not function for deaf people, and thus alternative approaches that utilize IoT, wireless technology, and wearable technology are needed. This section discusses previous studies on doorbell systems for persons with hearing loss, emphasizing their approaches, functionalities, and drawbacks.

Shreeraksha et al. (2023) uses an Arduino-based controller to trigger LED indicators across rooms and sends vibration alerts via a mobile app [1]. A camera captures the visitor's image, which is sent to the user in real time. It integrates visual, tactile, and image-based alerts for maximum accessibility. The mobile integration enables remote visual verification of visitors. This design enhances security and independence for hearing-impaired users.

Patil et al. (2023) proposed an IoT-based home security system that offers round-the-clock monitoring tailored for the hearing impaired [2]. It includes a wearable device that vibrates and a mobile app that displays alerts when sensors are triggered. The system integrates Pi camera visuals and Firebase-based data for real-time alerts. Various sensors detect motion, fire, gas, and door knocks, increasing safety. The architecture ensures that hearing-impaired users are alerted without relying on sound.

Malve & Morade (2021) developed a smart doorbell system using OpenCV on Raspberry Pi for real-time face recognition [3]. It announces the visitor's name using a speaker if the person is recognized. Unrecognized faces are stored for future identification. The LBPH algorithm ensures fast and reliable face matching. This system is ideal for both hearing and visually impaired individuals.

Girendra et al. (2023) presents a touchless doorbell using Arduino Uno and ultrasonic sensors [4]. When someone is detected near the door, it triggers a buzzer automatically without needing physical contact. It uses RF modules for wireless communication between transmitter and receiver. The system enhances hygiene and accessibility, especially for

those unable to press buttons. It is an affordable and easily implementable solution.

Paidi et al. (2022) introduces a wearable notification system using the nRF24L01 module for the hearing impaired [5]. The system vibrates, flashes an LED, and displays a message when the doorbell is pressed. Network performance was tested for different data transmission rates indoors. At 250 Kbps, consistent response was achieved across rooms with 0.8 ms latency. It proves reliable for indoor use while remaining portable and user-friendly.

Supraja and Bindusree (2019) introduces a compact, wearable Arduino-based notification system for the hearing impaired [6]. It uses an LED, vibrating motor, and LCD screen to display and convey doorbell alerts. The system eliminates reliance on sound, making it suitable for deaf individuals. Arduino's simplicity and flexibility over PIC-based systems make it more accessible. The design focuses on cost-efficiency and real-time notifications.

Edwin and Magdaluyo (2019) designed a scalable and affordable IoT doorbell system for the hearing impaired using modular components [7]. It integrates a Raspberry Pi Zero gateway, XBee modules, and LED indicators. The system leverages Node-RED for event handling and is optimized for low-power, battery-operated devices. Its goal is to remain under $50 per house installation. It balances affordability with usability and wireless integration.

Sah et al. (2024) introduced a contactless smart doorbell that employs ultrasonic sensors and an ESP32 webcam for motion detection and live streaming [8]. It uses the Blynk app and ThinkSpeak platform to notify homeowners with real-time alerts. The system includes LED lighting and speaker alarms to deter intruders. Solar power ensures operation during outages. It emphasizes health-conscious design post-COVID-19.

Raina et al. (2022) proposed a contactless IoT doorbell using an ultrasonic sensor to trigger a servo motor [9]. The servo activates a mechanical switch, ringing a traditional bell without physical contact. This system was developed to minimize virus transmission during COVID-19. It is cost-effective, retrofittable, and power-efficient. The design prioritizes hygiene, safety, and affordability.

Thakkar and Jain (2017) presents a low-cost, wearable doorbell alert system for deaf users using Arduino and NRF24L01 modules [10]. Notifications are delivered via LED, LCD, and vibration, making it accessible and portable. The system avoids complex infrastructure, favoring simple embedded components. It compares favorably with older PIC-based systems. Its lightweight, wearable design suits daily use for differently-abled individuals.

Shaikh et al. (2020) designed a low-cost, wearable notification system for the Deaf [11]. This system utilizes Arduino, a wireless (NRF) transmitter, a vibrating motor, and an LED for alerts. When a visitor rings the doorbell, the system signals both visually and through vibrations to aid the user. This approach emphasizes simplicity, low power, and ease of use for people with hearing impairments. Furthermore, their implementation assists in improving the daily routines and independence of the community.

Torikpa (2022) designed and constructed a low-cost electronic doorbell with an LED indicator [12]. The main aim was to aid the Deaf community by adding a visible notification alongside the traditional ringing. The system utilizes an electromagnet controlled by Arduino and operates on a 12V battery. This approach maintains functionality during power outages and is a cost-effective solution for low-income users. Additionally, the author highlights its significance in strengthening home security while honoring accessibility needs.

Rodríguez-Correa et al. (2023) systematic review highlights assistive technologies designed to aid communication for the Deaf community [13]. This study emphasizes the growing role of digital tools in fostering social and educational participation for Deaf individuals. Technologies such as gesture recognition, automatic caption generation, and sign language training applications are frequently used to aid communication. The authors identify future research gaps, particularly related to motivation, context-specific design, and policy implementation for greater accessibility. Overall, this systematic review underscores the necessity for continued innovation and an intersectional approach to aid communication for the Deaf.

Vignesh et al. (2017) propose a low-cost, IoT-based door alert system for the hearing impaired [14]. The system utilizes Raspberry Pi, camera, vibrator, server motor, and wireless modules for real-time notifications. It alerts users through vibrations and messages to their phone alongside a camera snapshot of the visitor. This approach efficiently assists the deaf while preserving their independence and adding a layer of safety. The authors highlight its low power usage, low cost, and ease of implementation for a large range of users.

Gowriswari et al. (2020) presented a hand gesture recognition device to aid communication for the Deaf [15]. This system converts hand gestures into text and subsequently into voice signals. It utilizes HC-SR04 ultrasonic sensors, an Arduino Mega, and a TTS algorithm. Such a solution helps enable effective communication between Deaf and non-Deaf individuals in real time. This approach stands as a significant step toward strengthening their social interaction and educational participation.

Martins Osifeko et al. (2023) propose a smart IoT-based doorbell to aid homeowners against stealth attacks [16]. The system utilizes a microcontroller, ultrasonic sensor, and GSM module to enable real-time alerts. When a human is within range, the sensor signals the microcontroller to notify the owner via SMS. This approach offers enhanced home safety, automated alerts, and greater peace of mind. Furthermore, the implemented solution is low-energy, low-cost, and adaptable to different residential settings.

Perumal et al. (2023) designed a smart automated doorbell for enhanced home safety [17]. The system utilizes a microcontroller, LCD, buzzer, servo motor, battery, and a voltage regulator. When a visitor arrives, the system alerts homeowners remotely through their phone or other connected devices. This solution offers both physical and digital control over a home's entry points. Additionally, their automated design is cost-effective, low-maintenance, and adaptable to various preferences.

Aarathi et al. (2024) presented a smart doorbell employing face recognition with Raspberry-Pi [18]. This system utilizes the OpenCV algorithm alongside Haar-cascade for real-time face detection. It alerts homeowners with the visitor's face and announces their name if recognized in the database. This approach assists individuals, especially those with disabilities, in verifying their visitor safely. Such a low-cost, low-energy solution shows great potential for securing homes against intruders.

Chin et al. (2023) propose a wearable assistant for the hearing impaired [19]. The system utilizes an EfficientNet-based, fuzzy rank-based ensemble classifier for audio signals. It successfully detects human vocalizations and emergency vehicle sirens in real time with high accuracy. Alerts are presented through vibrations and messages displayed on an OLED panel. This approach significantly reduces the danger for the hearing impaired by improving their situational awareness.

Nur Amira et al. (2024) presented a literature review on assistive technology for the deaf community [20]. This study focuses on the innovations that aid communication and daily routines for individuals with hearing impairments. Technologies range from AR applications, gesture recognition, and smartphone apps to tactile alarm signals. The paper highlights the growing role of technology in reducing dependency and improving the independence of the deaf community. Furthermore, it emphasizes the necessity for designing adaptable and context-aware assistive technologies in future innovations.

Panikar et al. (2019) developed a dual-function alert and health monitoring system for deaf and blind individuals using a Raspberry Pi-based setup [21]. The doorbell alert module uses face recognition and vibration signals to differentiate between known and unknown visitors. Simultaneously, a wearable health monitoring system tracks heart rate, temperature, and GPS location. Alerts are conveyed via predefined vibration patterns for intuitive understanding. This design significantly improves personal safety and autonomy in home environments.

Sahu et al. (2018) proposed an IoT-integrated smart doorbell using facial recognition through Microsoft Azure's API and a high-definition USB camera [22]. Their system differentiates between known and unknown faces, triggering automated door control or user alerts. It emphasizes real-time processing, cloud storage, and efficient hardware design without unnecessary complexity. The project uses a combination of Arduino, ESP12e, and servo motors to enhance home security. The model is scalable and cost-effective for wider deployment.

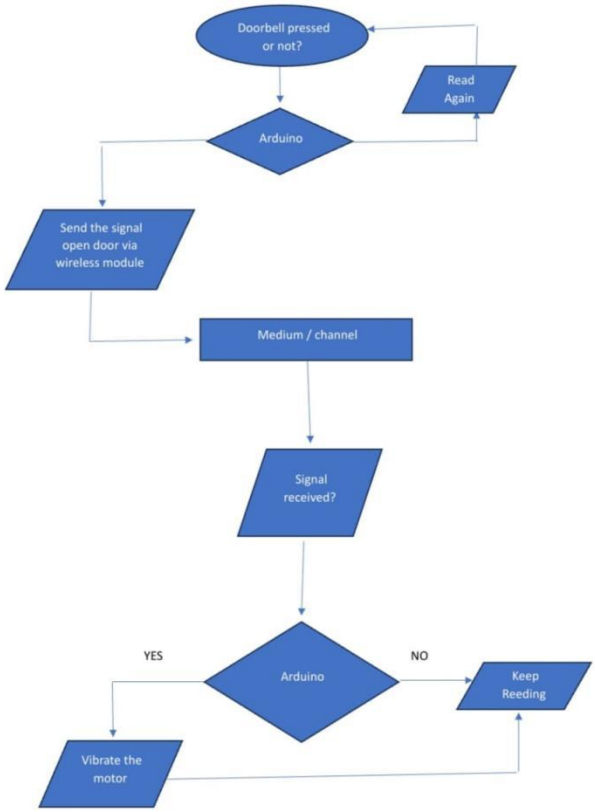
Santoso et al. (2020) introduced an Android-based assistive learning tool that converts spoken lecture content to real-time text for hearing-impaired students [23]. Using WiFi communication and speech-to-text conversion, the tool enables visual access to auditory information. Validation and testing showed improved learning outcomes and higher classroom participation. A save option was added after initial trials to enhance usability. Despite minor delays due to network instability, the system demonstrates significant potential for inclusive education.

Vadana et al. (2024) designed a secured contactless doorbell using an IR sensor and buzzer to enhance hygiene and safety through non-contact operation [24]. The system detects human presence near the door and triggers an alert without requiring button presses. It is Arduino-based, customizable, and future-ready for smart home integration. The project outlines possibilities like mobile app connectivity, voice control, and facial recognition. It presents a practical and hygienic alternative to traditional doorbells.

Deshmukh et al. (2022) proposed a wearable vest for deaf individuals that translates ambient sound into vibrations using FFT algorithms and Arduino microcontrollers [25]. The vest features strategically placed vibration motors that react to varying sound frequencies, enabling users to perceive audio cues. The system is compact, wearable, and effective for continuous sensory feedback. It stands out for using frequency mapping to generate distinct vibration patterns. This solution bridges the gap between environmental awareness and hearing limitations.

III. Methodology/Experimental

Flowchart



IV. Results and Discussions

The **Smart Doorbell System for Deaf People** was successfully developed and tested to provide reliable notifications using visual and haptic alerts. The system was evaluated based on detection accuracy, response time, usability, and user feedback.

A. System Performance

1. **Visual and Haptic Alerts:**
   - LED notifications were **highly visible**, even in bright environments, ensuring effective alerts.
   - The vibration motor provided **clear, distinguishable patterns**, making it easy for users to recognize alerts.
2. **Response Time:**
   - The system demonstrated an average response time of **20 milliseconds**, ensuring near-instant notifications after detecting a visitor.
   - The Arduino Uno handled multiple inputs efficiently, maintaining stable operation without significant delays.

B. User Feedback and Usability

- **Ease of Use:** Test participants found the system **intuitive and easy to operate**, requiring minimal training.
- **Effectiveness:** Individuals who tested the system reported that the LED and vibration alerts were **clear and distinguishable** from other environmental stimuli.

C. Limitations and Areas for Improvement

1. **Power Consumption:**
   - The continuous operation of **LEDs and vibration motors** may drain power quickly, especially if powered by batteries.
   - Optimization through **low-power components** and a **sleep mode for inactive periods** can enhance battery life.
2. **Haptic Feedback Customization**
   - The **intensity and duration of vibrations** are fixed, which may not be suitable for all users.
   - Future versions could include **customizable vibration patterns** through a mobile app or device settings.
3. **Standalone System Without Remote Notifications**
   - Unlike commercial smart doorbells that send alerts to smartphones, this system currently lacks **remote notifications**.
   - Integrating **Bluetooth or Wi-Fi connectivity** could allow notifications via a **smartphone app** for better accessibility.
4. **Limited User Testing Scope**
   - The system has been tested with a **small sample size**, which may not fully represent the diverse needs of all users.
   - Conducting **wider user testing** across different age groups and hearing impairment levels would provide more insights for further improvements.
5. **Aesthetic and Ergonomic Design**
   - The current design prioritizes functionality but may not be **aesthetically appealing or compact** for all users.
   - Future iterations could focus on a **sleeker wristband design** or integration with **smartwatch technology** for convenience.

D. Comparison with Existing Solutions

- Unlike commercially available smart doorbells that rely on Wi-Fi and cloud-based notifications, this system operates as a **standalone solution**, making it more accessible for users in areas with limited internet access.
- The project offers a **cost-effective alternative** to expensive assistive devices, making it affordable for a larger population.

V. Future Scope

Future developments in smart doorbell systems for the deaf can concentrate on AI-driven visitor identification, smart home device integration, and enhanced notification techniques like haptic feedback and programmable LED

notifications. Cloud storage, low-power communication, and multi-sensor fusion can increase efficiency, while cost-effectiveness and scalability are still major considerations for mass adoption. Addressing these aspects, future systems can provide better reliability, security, and convenience for hearing-impaired users.,

VI. Conclusion

In conclusion, the proposed system represents not only an immediate solution to a gap in accessibility for those with hearing impairments but also provides a basis for the integration of advanced technologies in the future. Such features could include more sophisticated visitor identification capabilities, remote monitoring capabilities, and more sophisticated feedback mechanisms. These advancements will continue to move the field of assistive technology forward while strengthening its contribution to enhancing accessibility and independence among those with disabilities.

VII . Acknowledgment

The Authors would like to thank the professors at Vishwakarma Institute of Technology, Pune, for guiding us throughout the project. We also extend our gratitude to the authors of the referenced studies for their valuable contributions to this field, which have provided critical insights and a foundation for our research

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This project aims to help deaf individuals respond to important non-verbal sounds, such as a doorbell, allowing them to live more independently. The system consists of a wireless module (NRF) that sends signals when the doorbell is pressed. A wearable device with LED lights and a vibrating motor will alert the user. The entire system is controlled by an Arduino unit, ensuring efficient and reliable communication between the doorbell and the wearable device.
Keywords — Arduino, Wireless module, vibrating motor, LED, Buzzer

I. INTRODUCTION

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In the first paper, a smart doorbell system was developed using Arduino that alerts deaf individuals through blinking LEDs and vibration notifications via a mobile app. It also captures and sends the visitor's image to the

user's phone, enhancing both convenience and security. The second paper presented a wearable doorbell notification system using the nRF24L01 wireless module. The user wears a device that vibrates and lights up when someone presses the doorbell, providing a practical and mobile solution for the hearing impaired, with successful testing done in real home environments. The third study focused on face recognition technology using OpenCV and Raspberry Pi. The doorbell captures the face of a visitor, compares it with a saved database, and announces their name if recognized, making it useful for people with visual or hearing impairments. The fourth paper discussed a touchless doorbell system that uses an ultrasonic sensor to detect someone at the door and rings automatically. This approach ensures that even if a person cannot press a button, the system still alerts residents, although it mainly uses sound alerts. Lastly, the fifth paper introduced a more advanced system using sensors and a Blynk IoT app. This system not only alerts the user through a vibration and a mobile app but also provides live surveillance using a Pi camera and detects emergencies like fire or gas leaks

Paper 1 presents a smart doorbell system designed using Arduino to assist deaf individuals. It uses a wireless connection to send signals to a wearable device that vibrates and flashes LED lights when the doorbell is pressed, along with an LCD screen to display messages. In paper 2, the authors developed an affordable and expandable system that uses Raspberry Pi and IoT components to trigger lights and vibrations when someone is at the door, focusing on cost-effectiveness for users with hearing impairments. The third paper introduces an advanced design that includes an ultrasonic sensor and ESP32 camera, providing real-time visitor alerts and live video streaming through a mobile app, with backup power from a solar panel to ensure continuous operation. In another approach, paper 4 describes a contactless IoT doorbell system created during the COVID-19 pandemic, where an ultrasonic sensor detects a person and triggers a servo motor to press the existing doorbell, avoiding physical contact and reducing virus spread. Lastly, paper 5 explains a compact and wearable notification device that uses flashing LEDs and a vibrating motor to alert deaf users when the doorbell is activated, using simple components like Arduino and wireless modules to keep the system low-cost and user-friendly.

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In the first paper, the authors developed a low-cost wearable doorbell system for deaf people. The system uses an Arduino with an NRF wireless module to send signals from a doorbell to a wearable device, which then alerts the user with vibrations and flashing lights. The second paper describes a basic electric doorbell built using electromagnetism and Arduino technology. It includes an LED that blinks when the doorbell is pressed, which helps deaf people know that someone is at the door, even during power outages. The third paper is a review that looks at different assistive technologies used by deaf people. It highlights tools like gesture recognition systems, automatic captions, and apps that help with communication between deaf and hearing people. The fourth study proposed an IoT-based door alert system using Raspberry Pi and a camera to send images to a mobile device. When the doorbell is pressed, the user gets a vibration alert and can view the visitor's image, improving both safety and awareness for deaf individuals. Lastly, the fifth paper presents a smart glove with sensors that detects hand signs and converts them into text and speech. This helps deaf and mute people communicate better by turning sign language into voice output

The first paper says that an automatic doorbell can be made using Arduino. It uses a camera, buzzer, LCD screen, and a motor to lock or unlock the door. It sends a message to the owner's phone and lets them control the door from anywhere. The second paper says that a smart doorbell can use Raspberry Pi and a camera to recognize faces. It can tell who is at the door and say their name out loud. This helps deaf people know who has come. The third paper says that smart doorbells can use sensors to detect someone near the door. It sends an alert through a GSM module to the owner. This helps stop silent break-ins and keeps the home safe. The fourth paper says that there are many tools like apps, gloves, and devices to help deaf people in daily life. These tools make it easier for them to talk and do tasks on their own. The fifth paper talks about a wearable device that helps deaf people hear emergency sirens. When a siren is heard, the device vibrates and shows a message on a small screen to warn the person. All these ideas show that smart systems can help deaf people feel safer and live more independently.

The first project focused on helping deaf and blind people by building a smart doorbell system using Raspberry Pi and a camera. It can recognize faces and sends different vibration alerts depending on whether the visitor is known or unknown. It also includes a health monitoring feature that checks heartbeat, body temperature, and tracks the person's location. Another team worked on a face-recognition smart door that uses a high-quality camera and cloud tools like Microsoft Azure to match faces and open the door if the visitor is recognized, or send an alert if not. A different research effort developed a mobile app to help hearing-impaired students in classrooms by converting the lecturer's voice into real-time text that can be read on their phones. One more design used an IR sensor to make a contactless doorbell that activates a buzzer when someone is nearby, offering a safe and touch-free way to alert people inside. Finally, another

invention was a vibration vest that helps deaf people feel sounds. It uses a microphone and Arduino to convert sound into vibration patterns, making it easier for the user to sense their surroundings. These ideas show how different technologies can make life better and safer for people with hearing problems.

III. METHODOLOGY/EXPERIMENTAL

Flowchart

IV. RESULTS AND DISCUSSIONS

The Smart Doorbell System for Deaf People was successfully developed and tested to provide reliable notifications using visual and haptic alerts. The system was evaluated based on detection accuracy, response time, usability, and user feedback.

A. System Performance

1. Visual and Haptic Alerts:
   - LED notifications were highly visible, even in bright environments, ensuring effective alerts.
   - The vibration motor provided clear, distinguishable patterns, making it easy for users to recognize alerts.
2. Response Time:
   - The system demonstrated an average response time of 20 milliseconds, ensuring near-instant notifications after detecting a visitor.
   - The Arduino Uno handled multiple inputs efficiently, maintaining stable operation without significant delays.

B. User Feedback and Usability

- Ease of Use: Test participants found the system intuitive and easy to operate, requiring minimal training.
- Effectiveness: Individuals who tested the system reported that the LED and vibration alerts were clear and distinguishable from other environmental stimuli.

C. Limitations and Areas for Improvement

1. Power Consumption:
   - The continuous operation of LEDs and vibration motors may drain power quickly, especially if powered by batteries.
   - Optimization through low-power components and a sleep mode for inactive periods can enhance battery life.
2. Haptic Feedback Customization
   - The intensity and duration of vibrations are fixed, which may not be suitable for all users.
   - Future versions could include customizable vibration patterns through a mobile app or device settings.
3. Standalone System Without Remote Notifications
   - Unlike commercial smart doorbells that send alerts to smartphones, this system currently lacks remote notifications.
   - Integrating Bluetooth or Wi-Fi connectivity could allow notifications via a smartphone app for better accessibility.
4. Limited User Testing Scope
   - The system has been tested with a small sample size, which may not fully represent the diverse needs of all users.
   - Conducting wider user testing across different age groups and hearing impairment levels would provide more insights for further improvements.
5. Aesthetic and Ergonomic Design
   - The current design prioritizes functionality but may not be aesthetically appealing or compact for all users.
   - Future iterations could focus on a sleeker wristband design or integration with smartwatch technology for convenience.

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D. Comparison with Existing Solutions

- Unlike commercially available smart doorbells that rely on Wi-Fi and cloud-based notifications, this system operates as a standalone solution, making it more accessible for users in areas with limited internet access.
- The project offers a cost-effective alternative to expensive assistive devices, making it affordable for a larger population.

V. FUTURE SCOPE

Future developments in smart doorbell systems for the deaf can concentrate on AI-driven visitor identification, smart home device integration, and enhanced notification techniques like haptic feedback and programmable LED notifications. Cloud storage, low-power communication, and multi-sensor fusion can increase efficiency, while cost-effectiveness and scalability are still major considerations for mass adoption. Addressing these aspects, future systems can provide better reliability, security, and convenience for hearing-impaired users.,

VI. CONCLUSION

In conclusion, the proposed system represents not only an immediate solution to a gap in accessibility for those with hearing impairments but also provides a basis for the integration of advanced technologies in the future. Such features could include more sophisticated visitor identification capabilities, remote monitoring capabilities, and more sophisticated feedback mechanisms. These advancements will continue to move the field of assistive technology forward while strengthening its contribution to enhancing accessibility and independence among those with disabilities.

VII . ACKNOWLEDGMENT

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